5 The first law of thermodynamics can be represented by the expression

$$\Delta U = q + w.$$

(a) State what is meant by the symbols in the expression.

.....

.....

..... [2]

(b) A fixed mass of an ideal gas undergoes a cycle $ABCA$ of changes, as shown in Fig. 5.1.

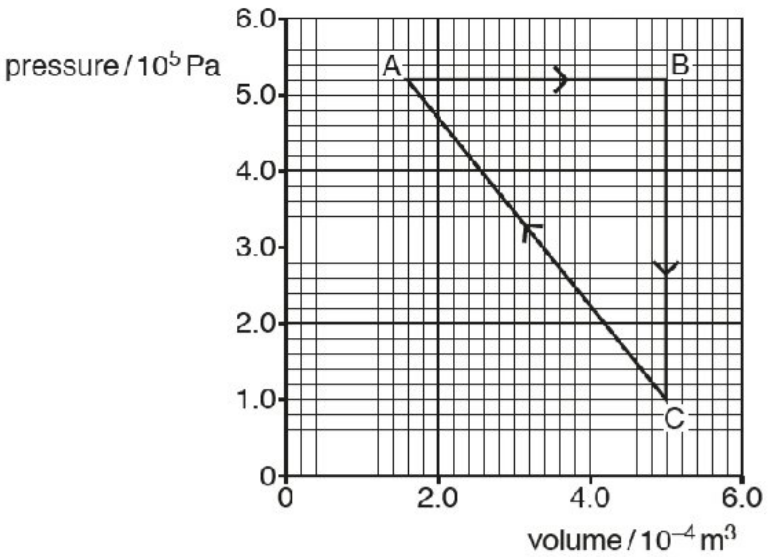


Fig 5.1

(i) During the change from A to B, the energy supplied to the gas by heating is 442 J. Use the first law of thermodynamics to show that the internal energy of the gas increases by 265 J

[2]

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(ii) During the change from B to C, the internal energy of the gas decreases by 313 J. By considering molecular energy, state and explain qualitatively the change, if any, in the temperature of the gas.

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..... [3]

(iii) Calculate the change in internal energy from C to A.

change in internal energy = ... J [1]

(iv) The temperature of the gas at point A is 227°C.

Calculate the number of molecules in the fixed mass of gas.

Number of molecules = ... [2]

[Total :10]

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